

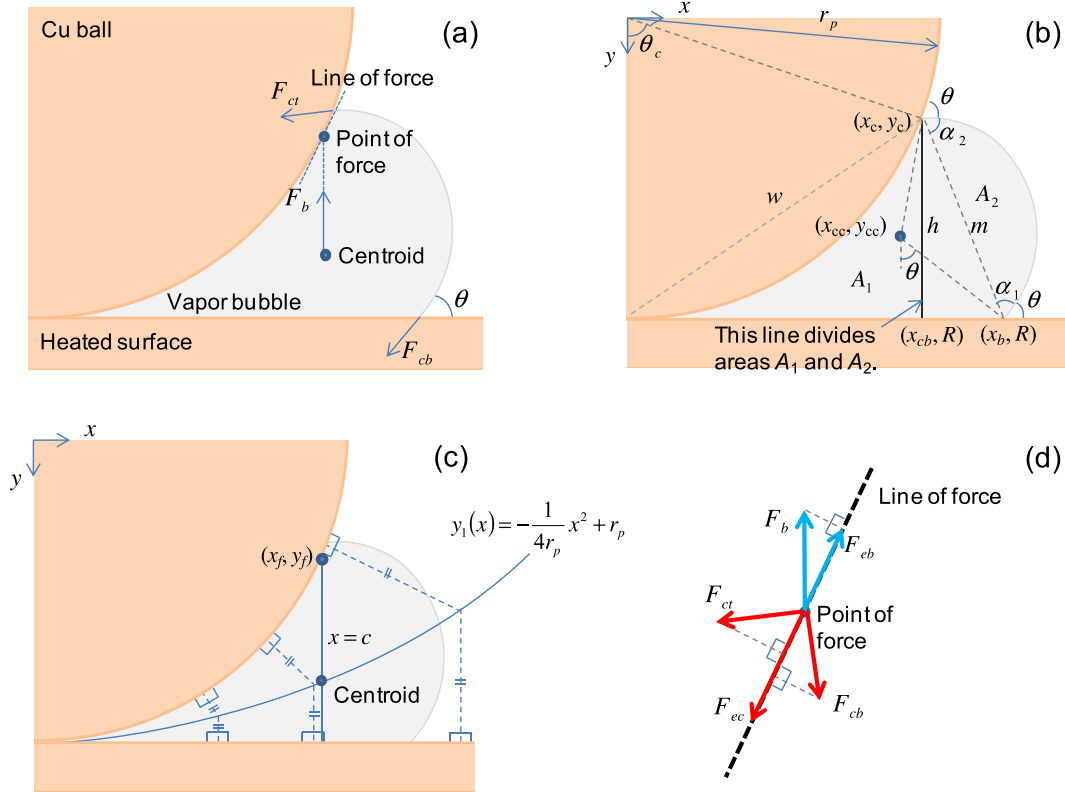


Fig. A3. Schematic diagram of the vapor embryo force balance model for a spherical particle on a heated surface. The model is used to determine the size of a vapor bubble required for release from the cavity. The diagram shows the (a) force balance diagram, (b) decomposed bubble geometry, (c) equation for centroid of the bubble, and (d) the effective buoyant force and capillary forces.

6. Project the buoyant and capillary forces onto the line of force and compare the effective buoyant force, F_{eb} , against the effective capillary force, F_{ec} .
7. Iterate steps 1 through 6 until the effective buoyant force equals

$$r_b = \frac{m}{2 \cos(|\pi/2 - \alpha|)} \quad (\text{A3})$$

$$\alpha = \alpha_1 - \alpha_2 = \cos^{-1} \left(\frac{h}{r_p} \right) + \alpha_2 - \pi \quad (\text{A4})$$